

NORILSK, Russian Federation

WMO: 230780

Lat: 69.3481N

Lon: 88.2597E

Elev: 65

StdP: 100.55

Time Zone: 7.00 (E07)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -45.3	(c) -43.2	(d) -47.3	(e) 0.0	(f) -45.0	(g) -45.4	(h) 0.0	(i) -43.1	(j) 18.5	(k) -11.1	(l) 16.7	(m) -19.3	(n) 2.9	(o) 130	(p) 0.903

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.5	26.5	17.7	23.9	16.3	21.2	15.2	18.2	24.3	17.1	22.7	15.9	20.7	3.8	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.9	11.4	20.4	14.7	10.5	19.5	13.5	9.7	18.4	51.7	24.2	48.4	22.4	44.9	20.6	21.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 14.3	(b) 12.0	(c) 10.4	(e) -46.2	(f) 28.0	(g) 2.3	(h) 2.4	(i) -47.9	(j) 29.7	(k) -49.3	(l) 31.1	(m) -50.6	(n) 32.4	(o) -52.2	(p) 34.1		
(4) 14.3	(5) 12.0	(6) 10.4	(4) -46.6	(5) 19.7	(4) 2.4	(5) 1.6	(4) -48.3	(5) 20.8	(4) -49.7	(5) 21.8	(4) -51.0	(5) 22.6	(4) -52.7	(5) 23.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-7.9	-27.4	-26.1	-16.9	-11.3	-3.1	10.1	15.4	11.1	4.6	-7.1	-22.0	-23.1		
	DBStd	16.59	10.15	9.46	9.44	7.23	5.45	5.55	4.84	4.12	4.40	7.09	8.24	9.61		
	HDD10.0	6847	1160	1011	834	638	405	67	10	36	169	531	959	1026		
	HDD18.3	9597	1418	1244	1092	888	663	252	118	227	411	789	1209	1284		
	CDD10.0	326	0	0	0	0	0	71	178	70	8	0	0	0		
	CDD18.3	35	0	0	0	0	0	6	27	2	0	0	0	0		
	CDH23.3	258	0	0	0	0	0	42	206	9	0	0	0	0		
	CDH26.7	43	0	0	0	0	0	6	37	1	0	0	0	0		
(14) Wind	WSAvg	4.9	5.5	5.2	5.7	5.5	5.0	4.4	3.8	4.0	4.0	4.8	5.1	5.7		
(15) Precipitation	PrecAvg	856	70	61	60	59	60	60	64	81	73	108	79	81		
	PrecMax	1131	136	109	157	121	100	110	120	142	150	176	142	148		
	PrecMin	531	15	16	11	6	21	14	6	28	33	32	16	10		
	PrecStd	125	32	25	30	27	21	23	31	26	26	36	33	32		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-6.4	-5.0	1.2	4.2	12.6	27.5	29.5	24.5	17.4	9.1	-2.5	-3.1		
		MCWB	-7.3	-5.9	-0.6	1.4	6.2	18.5	18.5	16.3	11.9	7.1	-3.4	-3.9		
	2%	DB	-9.4	-6.9	0.1	1.9	8.9	24.2	27.6	21.7	14.8	4.5	-4.3	-5.2		
		MCWB	-10.2	-7.7	-1.2	0.1	4.5	15.9	17.7	15.4	10.8	3.3	-5.0	-6.0		
	5%	DB	-11.7	-10.2	-1.4	0.6	5.8	21.4	25.9	19.6	12.9	2.7	-6.5	-8.0		
		MCWB	-12.3	-10.9	-2.8	-1.0	2.9	14.8	17.3	14.4	9.6	1.3	-7.3	-8.7		
	10%	DB	-14.0	-12.5	-3.5	-1.0	3.6	18.7	23.3	17.5	10.8	1.2	-10.4	-10.0		
		MCWB	-14.7	-12.9	-4.5	-2.4	1.5	13.3	16.5	13.6	8.2	0.1	-11.1	-10.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-7.3	-5.8	-0.2	1.6	6.9	19.5	20.3	17.6	12.8	6.8	-3.0	-3.7		
		MCDB	-6.3	-5.2	0.8	3.4	11.4	25.4	26.2	22.3	15.7	10.1	-2.8	-3.1		
	2%	WB	-10.1	-7.8	-1.4	0.4	4.7	17.4	18.6	16.4	11.3	3.4	-5.1	-6.0		
		MCDB	-9.3	-7.0	0.0	1.8	8.4	22.0	25.4	20.4	14.4	4.5	-4.4	-5.2		
	5%	WB	-12.3	-10.8	-2.8	-0.9	3.0	15.5	17.8	15.2	9.9	1.5	-7.3	-8.7		
		MCDB	-11.7	-10.2	-1.6	0.4	5.8	20.0	24.1	18.2	12.4	2.4	-6.6	-8.1		
	10%	WB	-14.5	-12.9	-4.6	-2.5	1.7	13.8	16.9	14.1	8.4	0.2	-11.0	-10.6		
		MCDB	-13.9	-12.5	-3.5	-1.1	3.5	18.5	22.8	17.0	10.7	1.2	-10.4	-9.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.4	6.9	7.1	7.8	5.9	7.4	7.5	6.2	5.0	5.1	6.2	7.1		
		MCDBR	7.2	7.4	6.0	8.4	8.3	11.5	10.6	9.5	7.7	5.0	7.0	7.3		
		MCWBR	6.9	7.0	5.2	7.7	5.6	6.0	4.2	4.8	4.7	4.1	6.6	7.1		
	5% WB	MCDBR	7.3	7.6	6.1	8.3	7.4	10.8	9.6	8.0	6.9	5.0	7.1	7.5		
		MCWBR	7.1	7.3	5.6	7.7	5.3	5.9	4.3	4.7	4.7	4.2	6.8	7.3		
(40) Clear-Sky Solar Irradiance	taub		0.228	0.262	0.320	0.347	0.386	0.397	0.426	0.417	0.344	0.265	0.215	N/A		
	taud		1.840	1.924	1.935	1.958	1.971	2.229	2.185	2.156	2.270	2.027	1.867	N/A		
	Ebn at Noon		13	434	645	763	779	791	745	694	653	427	11	0		
	Edh at Noon		5	49	97	129	144	116	118	108	72	42	4	0		
(44) All-Sky Solar Radiation	RadAvg		0.01	0.36	1.90	4.13	5.84	5.35	4.95	3.09	1.58	0.51	0.06	0.00		
	RadStd		0.00	0.05	0.21	0.35	0.44	0.46	0.74	0.35	0.16	0.09	0.01	0.00		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	(47) Regional (4 neighbors)	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A

Nomenclature: See separate page